

10	Answer: A
	Vertical equilibrium: $kx \cos \theta = mg$ --- (1) Vertical component of tension = weight Horizontal component of tension provides centripetal force $kx \sin \theta = mr \left(\frac{2\pi}{T} \right)^2$ --- (2) $\frac{(2)}{(1)} : \tan 27^\circ = \frac{4\pi^2 (0.108 \sin 27^\circ)}{T^2 (9.81)}$ $T = 0.62 \text{ s}$

11	Answer: A
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